



Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

ECEN 404

Status Update 1

Group #17: Patent Miner

List of team members:

Christian Casteel

CJ Vines

Kosi Ezewudo

Sponsor: Dr. Reddy Narasimha



Project Summary

Problem statement:

- The current keyword search implemented by the United States Patent and Trademark Office (USPTO) fails to capture contextual meaning, leading to inaccuracy of search results and expensive billing rates for litigation professionals.

Solution:

- Development of a patent database mining tool that optimizes the patent search process by retrieving contextually relevant patents faster and more accurately compared to a traditional database.

The image shows two side-by-side screenshots of a patent database interface. The left screenshot displays a list of search results for the keyword 'localization'. The right screenshot shows the full text of a selected patent document.

Search Results Table:

Date Publish...	Family ID	Pages	Title
25-09-25	10000078...	28	SCOPE TURRET
25-09-25	10000086...	91	COMPOSITIONS AND METHODS FOR DETECTION OF LIVER CANCER
25-09-25	10000086...	70	SOY-DERIVED BIOACTIVE PEPTIDES FOR USE IN COMPOSITIONS AND METHODS
25-09-25	10000086...	12	APPARATUS, SYSTEM, AND METHOD OF PROVIDING HAZARD DETECTION
25-09-25	10000086...	134	Interconnectable Tiling System
25-09-25	10000086...	40	Smart specialized brushing device and Systems and Methods
25-09-25	10000086...	22	BIOMETRIC AUTHENTICATION
25-09-25	10000086...	16	PERSONAL AUDIO ASSISTANT DEVICE AND METHOD
25-09-25	10000086...	64	SYSTEM AND METHOD TO HELP ENABLE CREATION OF, DISTRIBUTION OF, AND USE OF
25-09-25	10000086...	42	Organic-Inorganic Hybrid Bulk Assemblies and Methods
25-09-25	10000086...	10	SYSTEM AND METHOD FOR WATER TREATMENT INCENTIVE
25-09-25	62242490	24	Method and system for integratedly managing vehicle operation state
25-09-25	77859897	63	PEPTIDE
25-09-25	78134919	88	TREATMENT INVOLVING NON-IMMUNOGENIC RNA FOR ANTIGEN VACCINATION
25-09-25	78413837	30	METHOD FOR PRODUCING MILK LIKE PRODUCTS
25-09-25	78598778	648	GASTRIC INHIBITORY PEPTIDE RECEPTOR LIGANDS
25-09-25	80218437	95	LIVE ATTENUATED SARS-COV-2 AND A VACCINE MADE THEREOF

Document Viewer Details:

COMPOSITIONS AND METHODS FOR DETECTION OF LIVER CANCER

DOCUMENT ID	DATE PUBLISHED
US 20250297317 A1	2025-09-25

INVENTOR INFORMATION

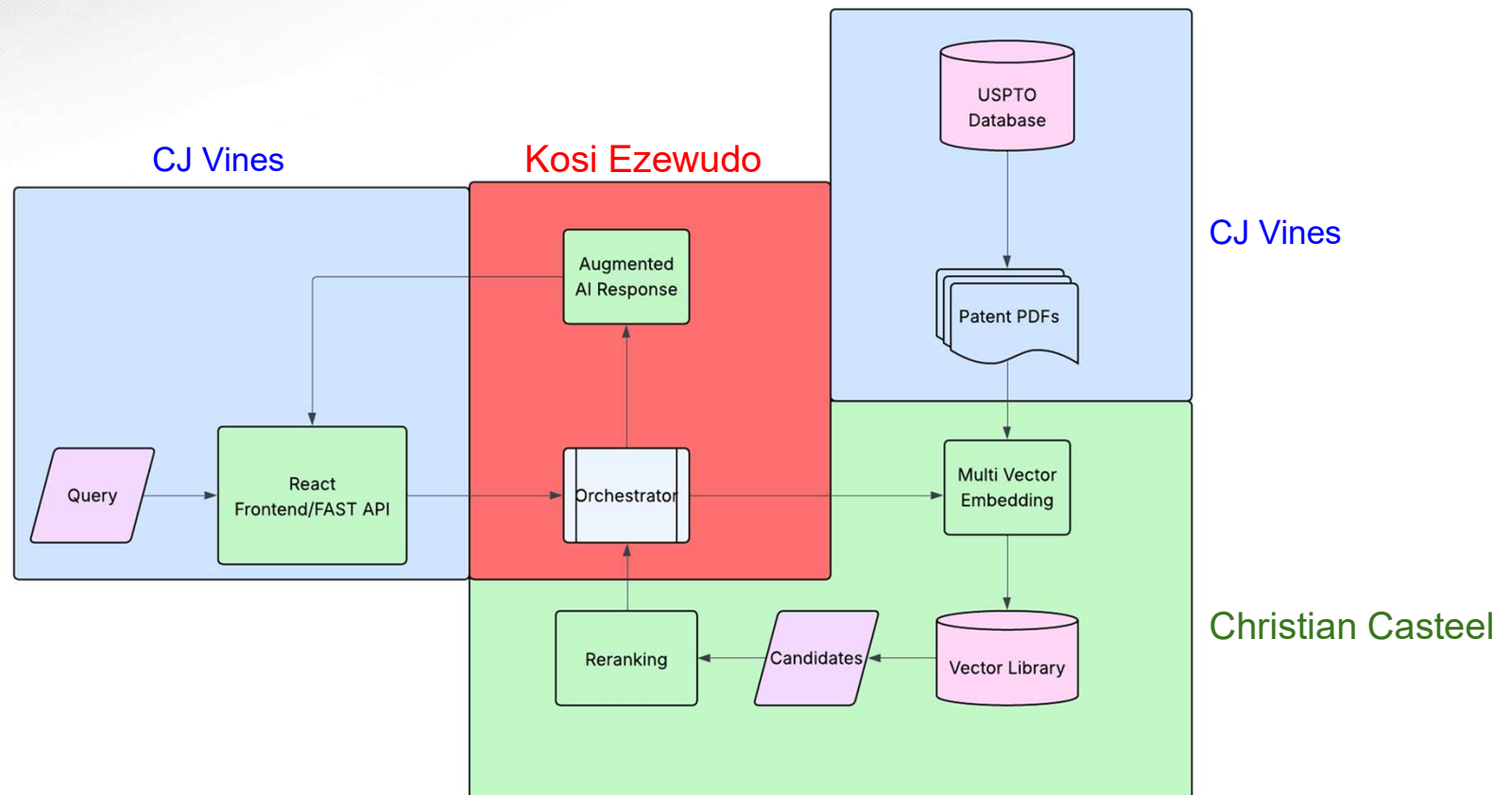
NAME	CITY	STATE	ZIP CODE	COUNTRY
Sedlak; Joseph Charles	Waltham	MA	N/A	US
Winn-Deen; Emily	Waltham	MA	N/A	US
Gusenleitner; Daniel	Waltham	MA	N/A	US
Couvillon; Anthony David	Waltham	MA	N/A	US
Bortolin; Laura Teresa	Waltham	MA	N/A	US
Salem; Deniel Parker	Waltham	MA	N/A	US
Biette; Kelly	Waltham	MA	N/A	US

APPLICATION NO 18/580422
DATE FILED 2022-07-21

US CLASS CURRENT: 1/1

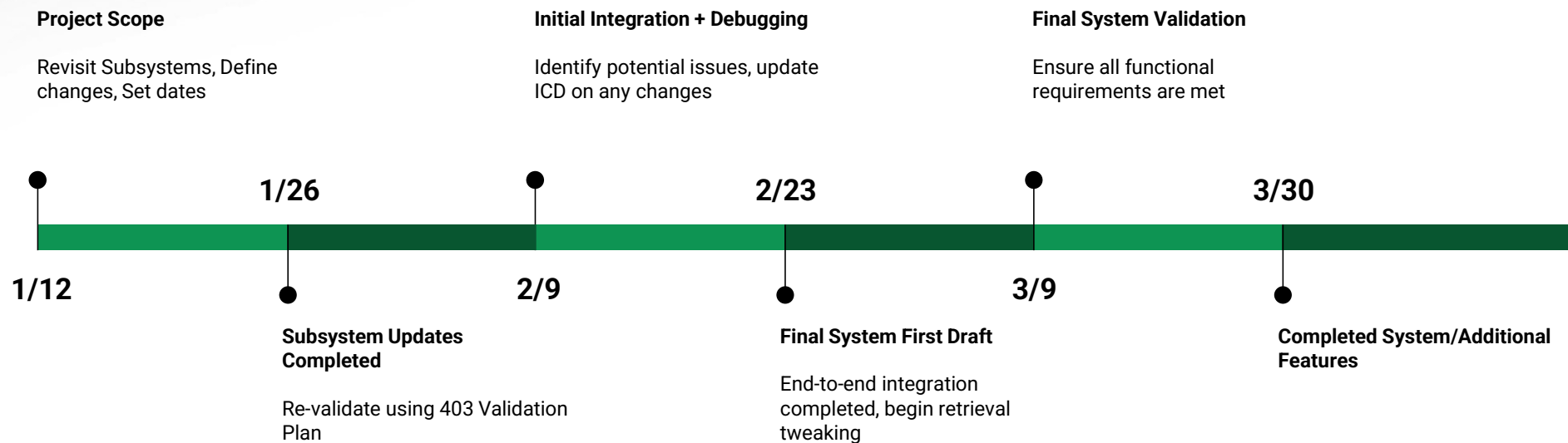


System Overview





Project Timeline





Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY

Major Project Changes

- Project scope: Patent Litigation Tasks
- Usage of ColBERT for patent re-ranking in addition to embedding
- Addition of MUVIRA to enhance retrieval speed and vector storage



Embedding Pipeline and Storage

Christian Casteel

Accomplishments since 403	Ongoing progress/problems and plans until the next presentation
• Created pipeline that embeds text from patents in XML format using ColBERT, and stores vectors locally using weaviate • Connected pipeline to USPTO download script	• Verify multi vector storage • Begin implementing ColBERT reranking



Embedding Pipeline and Storage

Christian Casteel

```
1 {
2   "deprecations": null,
3   "objects": [
4     {
5       "class": "PatentData",
6       "creationTimeUnix": 1763447837339,
7       "id": "001a7ba7-e896-41f4-9904-c641b2427f86",
8       "lastUpdateTimeUnix": 1763447837339,
9       "properties": {
10        "authors": [
11          "Gerard K. Cohen",
12          "Michelle D. Green",
13          "Christopher P. Smith",
14          "Craig Stuart",
15          "Kenneth L. Wright",
16          "Michelle M. Young"
17        ],
18        "chunk_index": 462,
19        "classification": "G06F21/31",
20        "content": "In some arrangements, the table 300 constitutes the digital fin",
21        "doc_id": "US1240603881",
22        "kind": "B1",
23        "priority_date": "2021-09-27",
24        "section": "description",
25        "title": "Systems and methods for location-binding authentication"
26      },
27      "vectorWeights": null
28    },
29    {
30      "class": "PatentData",
31      "creationTimeUnix": 1763447827756,
32      "id": "00767cf3-7715-4648-a113-09943f408f99",
33      "lastUpdateTimeUnix": 1763447827756,
34      "properties": {
35        "authors": [
36          "Ryan L. Roach",
37          "Cameron J. Day",
38          "Bryce W. Hobbs"
39        ],
40        "chunk_index": 256,
```

```
["deprecations":null,"objects":[{"class":"PatentData","creationTimeUnix":1763447837339,"id":"001a7ba7-e896-41f4-9904-c641b2427f86","lastUpdateTimeUnix":1763447837339,"properties":{"authors":["Gerard K. Cohen","Michelle D. Green","Christopher P. Smith","Craig Stuart","Kenneth L. Wright","Michelle M. Young"],"chunk_index":462,"classification":"G06F21/31","content":"In some arrangements, the table 300 constitutes the digital fingerprint. In such arrangements, the user device 110 can transmit the table 300 to a receiving device (e.g., the FI computing system 108, the backend authentication system 204, the computing device 202, etc.). In other arrangements, the table 300 is used by the user device 110 to generate a digital fingerprint. In such arrangements, the user device 110 transforms the information contained in the table 300 to create an alpha-numeric string that represents the information in the table 300 (e.g., by hashing the information contained in the table 300). In both arrangements, the digital fingerprint may be encrypted by the user device 110 prior to transmission to the receiving device.","doc_id":"US1240603881","kind":"B1","priority_date":"2021-09-27","section":"description","title":"Systems and methods for location-binding authentication"},"vectorWeights":null,"vectors":{"default":[0.2672888,0.124217324,0.1887971,0.034611937,0.33633468,0.15126854,-0.17636812,-0.21868779,0.19612898,-0.04731852,0.04651039,-0.13310894,-0.1490002,-0.1257096,-0.23580944,0.15466902,0.27875075,0.20804983,0.07802779,-0.02805733,0.18630764,0.14281929,0.11299232,0.1220121,-0.19977386,0.111223616,0.048800245,-0.3483961,-0.35408077,0.19103657,0.26236314,0.16105114,0.22627106,-0.14663775,-0.042216357,-0.033164263,-0.09030389,0.15663125,0.23715831,-0.57277256,-0.0016370378,-0.28937176,-0.17088543,0.16404423,-0.24251193,-0.043147072,-0.17419265,-0.033141356,-0.53299963,0.2987289,-0.52755207,0.22911802,0.15184255,0.2707046,0.3039899,0.17160863,-0.08858288,-0.15678754,0.05944864,-0.17791414,0.4044491,-0.07929333,0.15593426,0.24993464,0.20994359,-0.20469129,-0.121516265,0.11953705,0.17251743,-0.22973129,-0.062048085,-0.024363793,-0.083902664,-0.005580992,-0.251089,0.0038024057,-0.15419012,0.23779894,0.30620086,0.075820565,0.06345987,0.061227396,0.011808943,0.10166031,-0.046566825,-0.24071975,0.16698214,0.028742073,0.034761805,-0.3153073,-0.12681754,-0.18579215,0.24256623,0.10759292,-0.17754926,-0.19706179,-0.18470636,-0.21143328,0.011892267,0.042233407,-0.21363387,-0.030388119,0.118116826,-0.14474563,-0.036166247,-0.04534435,0.06604585,-0.011600691,-0.08113488,-0.0029605539,-0.12623422,-0.28585652,-0.12710676,-0.23260666,0.18912517,0.11025139,0.34533128,-0.020351734,0.13782473,-0.045392275,0.25364527,0.6550932,-0.08209082,0.04949695,0.15404575,-0.07572036,0.29823852,0.27595416,0.17552333,0.00079565594,-0.185627,0.34571224,0.18960825,0.05088066,0.063157976,-0.19986354,0.15496585,-0.063506395,0.12962128,-0.04775661,0.12664954,-0.20098266,0.41280618,-0.118681535,-0.20810625,0.11461531,0.1257581,-0.102201454,-0.3706633,0.17736326,0.052151877,0.2687811,-0.23837648,-0.22435844,-0.10444392,0.14719751,0.44389793,-0.12595974,-0.3585566,-0.060826622,0.06032641,-0.10871775,-0.043200348,0.093924224,0.27002835,-0.031589057,0.41514474,0.2509863,-0.04882544,0.10478484,-0.0405253,-0.33638987,-0.24305032,0.41934177,-0.03691494,0.18648991,0.19662043,0.12384446,0.0597624,-0.033739377,-0.2988496,0.29041296,-0.49323982,-0.046768975,-0.047013626,0.20877926,0.57603323,0.15969718,-0.087442026,0.057314884,-0.23984358,-0.69389796,0.35562268,-0.38325423,0.28868482,-0.29754356,-0.08794248,-0.24654087,-0.11762973,0.062621735,-0.046110567,0.1596731,0.28286707,-0.089381576,0.01231464,-0.06859363,0.014201247,-0.09227259,0.3403015,0.012319982,0.054863945,-0.123691484,-0.03492067,0.18856855,0.07697147,0.008112186,0.06790834,0.09589179,0.18531592,-0.11026927,0.11809165,0.24779774,0.31233534,0.211397,0.103732586,0.
```

Image of a patent chunk's metadata and a single embedding



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY

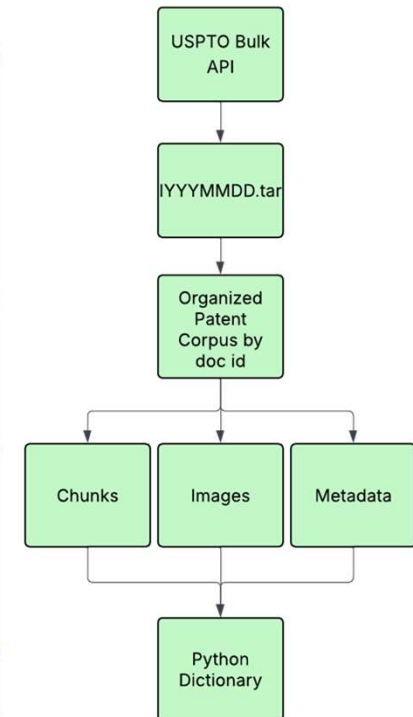
Frontend/Preprocessing CJ Vines

Accomplishments since 403	Ongoing progress/problems and plans until the next presentation
• Created a robust download pipeline with 100% retention of text and figures • Extracted all relevant metadata from patent, which is grouped with each chunk for filtered queries and LLM context • Created a React front end that successfully requests and displays relevant patents with hyperlinks to the full patent download	• Further refine SLURM job script for efficient usage of graphical and processing power • Assisting other subsystems in their implementation



Preprocessing

```
uspto_bulk_data > sampled > sample_05_12405102_Super-sensitive-capacitive-strain-sensor.md > abc # Super-sensitive ca
1  # Super-sensitive capacitive strain sensor with electrode fragmentation (Doc 12405102)
2
3  - Doc ID: 12405102
4  - Title: Super-sensitive capacitive strain sensor with electrode fragmentation
5  - Filing Date: 20210323
6  - Classification: G01B 7/22
7  - Authors: Gilles Lubineau; Hussein Nesser
8
9  ## Abstract
10
11 A capacitive strain sensor configured to measure a strain includes a dielectric layer;
12 face of the dielectric layer; and a second electrode placed on a second face, opposite
13 dielectric layer. The first electrode is formed of a single-walled carbon nanotube paper
14 nanotube paper has plural pre-cracks made according to a pattern.
15
16 ## Description
17
18 This application is a U.S. National Stage Application of International Application No.
19 23, 2021, which claims priority to U.S. Provisional Patent Application No. 63/002,850,
20 "HIGHLY SENSITIVE CAPACITIVE STRAIN SENSORS BASED ON FRAGMENTED ELECTRODES," the disclo
21 herein by reference in their entirety.
```



Sampled 50 random patents from each, reconstructed them in .md format, compared to pdf for accuracy



Frontend

Patents retrieved from patent chunks

US12403596-20250902-000001.png
Imaging process for detecting failures modes

US12403596-20250902-000002.png
Imaging process for detecting failures modes

US12403596-20250902-000003.png
Imaging process for detecting failures modes

US12403596-20250902-000004.png
Imaging process for detecting failures modes

US12403596-20250902-000005.png
Imaging process for detecting failures modes

US12403596-20250902-000006.png
Imaging process for detecting failures modes

Query
transistor

Answer (RAG)

(FAKE RAG) Proposed answer for: 'transistor'. See [1], [2], [3], [4], [5] for sources.

Based on the following 5 contexts:

[1] Imaging process for detecting failures modes: Various embodiments of the present technology generally relate to robotic devices, artificial intelligence, and computer vision. More specifically, some embodiments relate to an imaging process for detecting failure modes in a robotic mo...

[2] Switching power supply circuit and switching power supply system having the same: Disclosed herein is a switching power supply circuit that includes a switching circuit having an input node connected to an input power supply terminal and an output node connected to an output power supply terminal through an output swi...

[3] Systems and methods for location-binding authentication: Systems, methods, and apparatuses for authenticating a user based at least in part on a location of the user or a location of a user device are described. A method includes: receiving a login request including a user identifier associate...

[4] System and method for pruning neural networks at initialization using iteratively conserving synaptic flow: A system and method to prune parameters of a neural network at initialization uses iterative conserving synaptic flow that saves time, memory and energy both during training and at test time of the neural network. The result of the discl...

[5] Method for adjusting a phase of a carrier replica signal: In an example embodiment, a method for adjusting a phase of a carrier replica signal includes receiving, by a tracking loop, a digital baseband signal, generating within the tracking loop the carrier replica signal, generating within the the...

Refine your query...

Results

Cited Patents

[1] Imaging process for detecting failures modes

Various embodiments of the present technology generally relate to robotic devices, artificial intelligence, and computer vision. More specifically, some...

Title: Imaging process for detecting failures modes

Authors: Yide Shentu; David Mascharka; Tianhao Zhang; Yan Duany; Jasmine Deng; Xi Chen

Classification: B25J 9/1653

Filing Date: 2024/07/03

Kind: B2

doc_id: 12403596

[Google Patents full text](#)

[2] Switching power supply circuit and switching power supply system having the same

Disclosed herein is a switching power supply circuit that includes a switching circuit having an input node connected to an input power supply terminal and an...

Title: Switching power supply circuit and switching power supply system having the same

Authors: Saiki Shigemori; Akihiro Unno

Classification: H02M 3/158

Filing Date: 2023/08/07

Kind: B2

doc_id: 12407255

[Google Patents full text](#)

[3] Systems and methods for location-binding authentication

Systems, methods, and apparatuses for authenticating a user based at least in part on a location of the user or a location of a user device are described. A method...

Title: Systems and methods for location-binding authentication

Authors: Gerard K. Cohen; Michelle D. Green; Christopher P. Smith; Craig Stuart; Kenneth L. Wright; Michelle M. Young

Classification: G06F 21/31

Filing Date: 2021/09/27

Kind: B1

doc_id: 12406038

[Google Patents full text](#)

[4] System and method for pruning neural networks at initialization using iteratively conserving synaptic flow

Results

Cited Patents

[1] Imaging process for detecting failures modes

Various embodiments of the present technology generally relate to robotic devices, artificial intelligence, and computer vision. More specifically, some...

Title: Imaging process for detecting failures modes

Authors: Yide Shentu; David Mascharka; Tianhao Zhang; Yan Duany; Jasmine Deng; Xi Chen

Classification: B25J 9/1653

Filing Date: 2024/07/03

Kind: B2

doc_id: 12403596

[Google Patents full text](#)

[2] Switching power supply circuit and switching power supply system having the same

Disclosed herein is a switching power supply circuit that includes a switching circuit having an input node connected to an input power supply terminal and an...

Title: Switching power supply circuit and switching power supply system having the same

Authors: Saiki Shigemori; Akihiro Unno

Classification: H02M 3/158

Filing Date: 2023/08/07

Kind: B2

doc_id: 12407255

[Google Patents full text](#)

[3] Systems and methods for location-binding authentication

Systems, methods, and apparatuses for authenticating a user based at least in part on a location of the user or a location of a user device are described. A method...

Title: Systems and methods for location-binding authentication

Authors: Gerard K. Cohen; Michelle D. Green; Christopher P. Smith; Craig Stuart; Kenneth L. Wright; Michelle M. Young

Classification: G06F 21/31

Filing Date: 2021/09/27

Kind: B1

doc_id: 12406038

[Google Patents full text](#)

[4] System and method for pruning neural networks at initialization using iteratively conserving synaptic flow



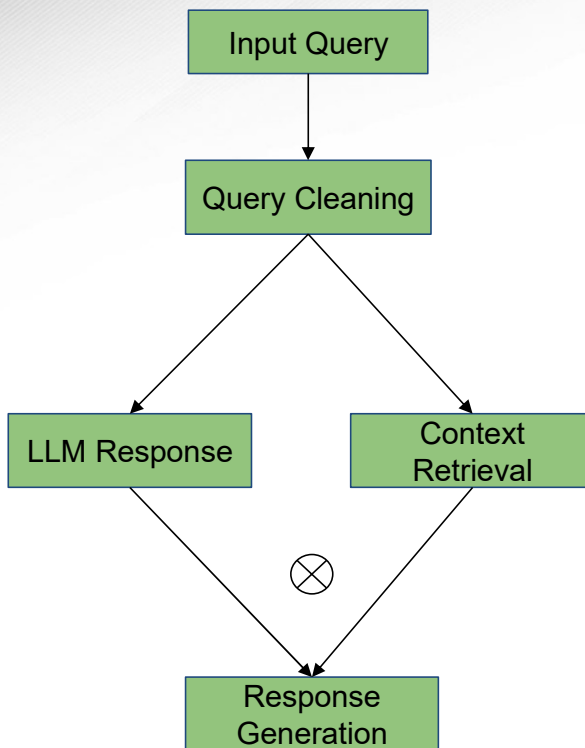
Retrieval Augmented Generation (RAG) Subsystem

Kosi Ezewudo

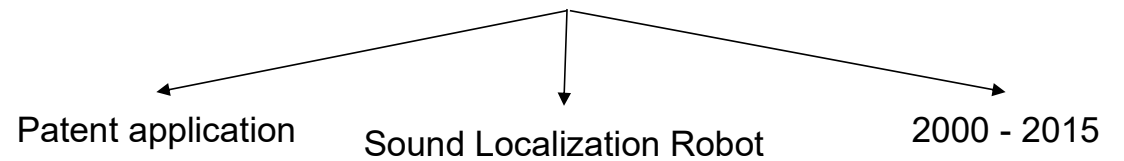
Accomplishments since 403	Ongoing progress/problems and plans until the next presentation
• Implemented a base RAG pipeline that; ◦ conditionally executes context retrieval • Implemented a RAG-layer that performs Query Cleaning • Created a stand-alone layer that performs Metadata Extraction	• Finalize and Integrate Metadata extraction layer to RAG pipeline • Finalize subsystem and demonstrate progress

Retrieval Augmented Generation Subsystem

Kosi Ezewudo



Sample Query: Are there any patent applications that were made specifically for sound localization robots between 2000 - 2015?



Execution Plan

[illegible]



Validation Plan

Paragraph #	Test Name	Success Criteria	Methodology	Status	Responsible Engineer(s)
3.2.1.2.	API/Pydantic Test	API gives correct response depending on the route taken 100% of the time	Send input query and data of different variations including edge cases like empty queries, queries with special characters, and get requests of the wrong shape	Tested	CJ Vines
3.2.1.1.	Determinism Test	Preprocessing Subsystem resulting chunks consistent across runs on the same dataset	Calculate SHA-256 hash of entire list of dictionaries after chunking and compare hash across runs to ensure download pipeline is deterministic. If any run is inconsistent, the test fails	Tested	CJ Vines
3.2.1.1.	Metadata Extraction	Stored metadata is verified to be at least 99% accurate	Compare metadata after chunking to metadata called from USPTO API, aiming for 99% accuracy	Tested	CJ Vines
3.2.1.4.	Full System Stress Test	Multiple simultaneous API calls are made successfully no data loss within 1ms	Make multiple API calls simultaneously, handle each concurrently in order.	Untested	CJ Vines & Kosi Ezewudo & Christian Casteel
3.2.1.5.	Embedding Reranking Test	"Golden" query - chunk pairs must have $\geq .05$ Recall@k, Mean Reciprocal Rank (MRR) $\geq .5$, and Normalized Discounted Cumulative Gain (NDCG) of .5	k = 10 target chunks at minimum with an equal number of matching queries. Calculate Recall@k, MRR, and NDCG for dataset. "Golden" set will include 50 hand selected patents.	Untested	Christian Casteel
3.2.1.5.	Embedding Accuracy Test	For a set of chunks, Hit@k of 1 and precision at k above .5. Relevance determined by patent id	Query containing information from multiple patents with similar CPC codes will test retrieval. Calculate Hit@k and Precision@k on top k chunks where k = 10.	Untested	Christian Casteel
3.2.1.1.	Failure Compliance Test	Pipeline can take in incorrect patent data and continue embedding without failure. Also will flag patents with false data	5 fake patents containing empty sections and incorrect CPC codes will be sent through the pipeline.	Untested	Christian Casteel
3.2.1.6.	Metadata Integrity Test	All stored metadata must be 100% consistent with metadata listed on a patent	Select 50 patents, specifically patents with generally unique CPCs and numbers of authors, then compare all metadata fields to the same patents on the USPTO database	Untested	Christian Casteel
3.2.1.1.	Download and Embedding Pipeline Stress Test	A USPTO bulk dataset is downloaded, embedded, and stored without failure	A USPTO bulk dataset download is conducted via frontend download script. The bulk dataset should contain minimum 1000 patents.	Untested	Christian Casteel & CJ Vines
3.2.1.6.	Query Metadata Extraction	Extract all relevant metadata from user query, and use the metadata to filter patent search	Execute patent search using queries containing various metadata including prior date, patent ID, CPC etc.	Untested	Kosi Ezewudo
3.2.1.3.	Query Cleaning	Removes typos, sentence fragments etc. from user query	Execute patent search using grammatically incorrect queries	Untested	Kosi Ezewudo
3.2.1.4.	Generation Accuracy and Quality Test	Generated responses must have $\geq .8$ Answer relevancy scores, $\geq .8$ Diversity and Novelty	Query Patent Miner using golden set of patents. Generate responses and calculate Answer relevancy, Diversity and Novelty scores	Untested	Kosi Ezewudo & Christian Casteel & CJ Vines



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY



Thank you!